



PULSE War Room

Comprehensive Audit Report

UI / UX / Design / Strategic Value Assessment
Bain Senior Partner Perspective

Classification	CONFIDENTIAL — Internal Use Only
Date	March 28, 2026
Prepared For	Alexander Laker, Strategy VP
Methodology	Live browser testing + deep source code review (32 components)
Assessment Standard	Bain Digital / McKinsey QuantumBlack benchmark
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PULSE WAR ROOM

COMPREHENSIVE UX / UI / STRATEGIC AUDIT

Profit Pool Unified Landscape Simulation Engine

Prepared for: Alexander Laker

Perspective: Bain Senior Partner | DAX-40 CEO Pitch Readiness

Date: 28 March 2026

Classification: CONFIDENTIAL

Audit Methodology: Live browser testing + full source code review of 32 components (~12,000 lines)

EXECUTIVE SUMMARY

This audit evaluates the PULSE War Room dashboard across five dimensions: functional reliability, user experience, visual design quality, strategic value for FMCG executives, and pitch readiness for a DAX-40 CEO board presentation. The audit was conducted through live browser testing of the deployed Vercel instance and deep source code review of all 32 React components (~12,000 lines of TypeScript).

Overall Verdict

The PULSE War Room is a professionally engineered, visually polished dashboard with genuine strategic value. However, it is not yet ready for a Bain Senior Partner to present to a Henkel ExCo. Three blocking issues prevent this: (1) the backend API is broken on Vercel, making the app completely inaccessible, (2) critical security flaws expose invite codes and credentials, and (3) the UX deviates from the specification's "single war room view" principle by fragmenting content across tabs and competing modal panels.

Audit Scorecard

1. CRITICAL FINDINGS — MUST FIX BEFORE ANY DEMO

1.1 Backend API Completely Broken

The Vercel serverless function returns `FUNCTION_INVOCATION_FAILED` on all auth endpoints. The `/api/v1/health` endpoint returns status OK with `trend_count: 0` and `model_loaded: false`, confirming the data layer is non-functional. This means no user can log in, no data loads, and the War Room is entirely inaccessible.

Root Cause: The Vercel Python runtime likely fails during database initialization (Postgres/SQLite cold start). The auth module attempts to seed default users on first request, and this initialization is crashing.

Impact: TOTAL BLOCKER. No feature can be evaluated in production.

Fix: Debug Vercel function logs, ensure database initialization handles cold starts gracefully, add health check that validates full auth pipeline.

1.2 Auth Forms Fail Silently

When the backend returns an error, the login and registration forms show no error message to the user. The form simply remains on screen with no feedback. Users have no idea whether their credentials are wrong, the server is down, or the invite code is invalid. This was observed during live browser testing: clicking “Sign In” with valid credentials resulted in the form clearing and reappearing with no error banner.

Root Cause: The `handleSubmit` function catches errors but relies on the parent `useAuth` hook to propagate them via props. If the hook fails to update the error state (e.g., due to a network exception), the UI shows nothing.

Fix: Add explicit error state clearing at the start of `handleSubmit`. Add a generic catch-all error message: “Unable to reach the server. Please try again.” Add a loading spinner on the Sign In button during API call.

1.3 Security Vulnerabilities

2. USER EXPERIENCE AUDIT

2.1 Spec Deviation: Tab Navigation vs. Single War Room View

The CLAUDE.md specification explicitly states: “Eight separate tabs forces executive users to navigate... Instead: THE WAR ROOM (Single Page Application with Contextual Panels).” However, the current implementation uses a two-tab navigation model (War Room | Trends), requiring users to switch tabs to see trend details. This violates the core design principle.

Recommendation: Merge the Trends tab into the Overview as a collapsible section or integrate trend information into the contextual drill-down panels (click a heatmap cell to see its underlying trends).

2.2 Modal Explosion — Competing UI Layers

The War Room has 6+ overlay panels that can compete for screen space: `CategoryDetailPanel` (right slide-in, 420px), `SettingsPanel` (right slide-in, 420px, same position), `DelphiPanel` (full-screen modal), `SessionSnapshots` (right slide-in, 460px), `ExecutiveBriefing` (full-screen overlay), and `CategoryDeepDive` (full-screen modal). Multiple panels can render simultaneously, creating cognitive overload. Apple’s design principle of “one thing at a time” is violated.

Recommendation: Implement a panel stack manager that ensures only one side panel and one modal are active at a time. Close the previous panel before opening a new one.

2.3 Header Button Overload

The sticky header contains 8+ interactive elements in a single row: Tab buttons (2), Scenario buttons (5), Convergence pill, Iterations pill, Simulate button, Export icon, Delphi icon, History icon, Briefing icon, and Settings icon. For a CEO, this is cognitive overload. A Bain presentation would never have 12 buttons in the header.

Recommendation: Reduce to 4 primary actions: Simulate, Scenario Selector (dropdown), Export, and Executive Briefing. Move Delphi, Snapshots, and Settings into a “More” menu or gear icon submenu.

2.4 Missing Error Recovery and Loading States

API failures are silently swallowed (empty catch blocks). No skeleton loaders appear while data fetches. No toast notifications confirm successful exports or score updates. No timeout detection for hung requests. If the API goes down mid-session, the user gets stale data with no warning.

Recommendation: Add skeleton shimmer loaders for all chart components. Add toast notification system (success/error). Add network status indicator. Implement retry logic for transient API failures.

2.5 No Responsive Design

The dashboard uses fixed grid layouts (1440px max width) with no media queries. On iPad (where ExCo presentations happen), the right panel (420px) covers most content. Header buttons wrap badly on screens below 1200px. Touch targets are adequate (32-36px) but the header is too cramped.

Recommendation: Add responsive breakpoints for 1024px (iPad landscape) and 1366px (laptop). On tablet, collapse side panel to bottom sheet. Reduce header to icon-only on small screens.

3. VISUAL DESIGN AUDIT

3.1 What Works Well

- Dark mode default with deep navy (#0F172A) background — matches spec exactly
- Glassmorphism panels with backdrop-filter blur(20px) — Apple-grade polish
- Framer Motion animations with spring physics (damping: 30, stiffness: 300) — butter-smooth
- Consistent design tokens (T.bg, T.text, T.accent) across all components
- Diverging green/red heatmap colors with opacity-based intensity scaling
- JetBrains Mono for data values, Inter for UI text — correct typographic hierarchy

3.2 Design Issues

- Light theme on auth page, dark theme on War Room — jarring transition. Auth should use dark theme too.
- Color-blind accessibility gap: red/green diverging scheme is unreadable for ~8% of males. Add red/blue colorblind mode toggle.
- No loading shimmer/skeleton — spec explicitly mentions “loading shimmer” for smooth UX.
- Tooltip styling inconsistent across components (PathTimeline uses tooltipStyle function, ForceRadar hardcodes dark gray).
- Design token color leakage: some components hardcode rgba values instead of using T.border.
- Auth page uses white/light card design which contradicts the dark-mode-first specification.

4. STRATEGIC VALUE AUDIT

4.1 High-Value Components (Keep and Polish)

Executive Briefing — 9/10

5-slide guided storytelling interface that transforms raw shift matrix data into a board-ready narrative. Follows the classic Bain deck flow: Portfolio Overview → Risks → Opportunities → Causal Story → Recommendations. This is THE component for CEO presentations. Keyboard navigation, speaker notes, and dot indicators are production-quality. No issues found.

Delphi Panel — 8/10

Formal 3-round expert elicitation with inter-rater reliability (Krippendorff's alpha). This is how PULSE earns credibility: structured scoring beats gut estimates. Card-by-card wizard flow is intuitive. Slider with semantic labels (Negligible to Transformative) is well-designed. Missing: estimated time-to-complete and explicit highlighting of low-agreement trends in Round 3.

Causal Flow (DAG Visualization) — 8/10

Custom SVG hexagonal layout showing force interdependencies with BFS shock propagation. Click a force to shock it, watch purple propagation cascade through the DAG. This is exactly what a strategy partner wants: show how a regulatory shock (Government) cascades through Technology → Customer → Consumer. Visually striking. Missing: configurable shock magnitude (currently hardcoded at +30%) and multi-round equilibrium visualization.

Heatmap — 7/10

Category × Year heatmap (not Category × Force as specified). Actually a better UX choice — shows continuous paths directly in the grid. Robust data extraction handles 5 different backend schema formats. Click to select, double-click for deep dive. Missing: keyboard accessibility for tooltips, and Force × Category toggle as specified.

4.2 Gaps vs. Specification (Missing Strategic Features)

4.3 Scope Creep Assessment

Several components add complexity without proportional CEO-level value. These should be deferred to Phase 2:

- EmergingTrends (1,012 lines): 19-API scanner with complex orchestration. Valuable for analysts but not for a CEO demo. High maintenance burden.
- SessionSnapshots (875 lines): Audit trail is compliance-valuable but not visually compelling. Defer to Phase 2.
- ProductImpactRankings (637 lines): Hardcoded 55-item product type map. Useful for category VPs, not ExCo. Move to drill-down view.

Removing these three components would reduce codebase by ~2,500 lines and sharpen the MVP to its core strategic value.

5. AUTH PAGE DEEP DIVE

5.1 Registration Flow Bug: Invite Code Pre-filled

When a user clicks “Register,” an “Invite Code” field appears pre-filled with “PULSE-2026.” This is both a UX failure and a security failure. The invite code exists as an access gate, but pre-filling it means anyone who visits the page can register. The field should be empty with a placeholder like “Enter your invite code” and should be masked (password-type input).

5.2 Tab Switching State Leak

When switching from Register to Sign In, the form retains values from the Register tab (Name, Invite Code fields). In live testing, after filling out the Register form and switching to Sign In, the Name and Invite Code fields appeared overlaid on the Sign In form. The `switchMode()` function clears error states but does not clear form field states (`name`, `inviteCode`).

Fix: Reset all form fields in `switchMode()`: `setName("")`, `setInviteCode("")`, `setEmail("")`, `setPassword("")`.

5.3 Forgot Password Flow

The Forgot Password link exists and leads to a reset token flow. However, the reset token deep-linking (via URL hash) is implemented without server-side email delivery. The password reset success state has a 3-second auto-redirect. Token expiration errors lack a recovery path (no “request new link” button).

6. COMPONENT QUALITY SCORECARD

Scores are 0–10. Design = visual polish. UX = user flow quality. Code = TypeScript quality, error handling. Value = strategic value for CEO pitch. A11y = accessibility. Tests = test coverage (0 across the board — no Jest/Vitest tests found anywhere).

7. PRIORITIZED RECOMMENDATIONS

7.1 P0: Blocking — Fix Before Any Demo

- Fix Vercel backend API. Debug `FUNCTION_INVOCATION_FAILED`. Ensure database initialization handles cold starts. Add retry logic. This is the single most important fix.
- Remove invite code pre-fill. Empty the field, mask it as password-type input. Move invite codes to environment variables only (not in source code).
- Remove hardcoded credentials from source. The default admin password should be generated dynamically or set via environment variable. Never in source code.
- Add error feedback on auth forms. Show error banners when login/register fails. Add loading spinner on submit button. Clear errors on new submission attempt.

7.2 P1: High — Fix Before CEO Pitch

- Add 80% confidence interval to headline KPI (p10/p90 bounds next to point estimate)
- Merge Trends tab into Overview (collapsible section or contextual panel)
- Reduce header to 4 primary buttons (Simulate, Scenario, Export, Briefing)
- Add skeleton loaders for all chart components during data fetch

- Implement panel stack manager (one side panel + one modal at a time)
- Add velocity visualization and early-warning trigger markers on PathTimeline
- Fix AllocationChart risk calculation to use copula-aware covariance, not naive variance

7.3 P2: Medium — Polish for Production

- Add responsive breakpoints for iPad (1024px) and laptop (1366px)
- Add colorblind-safe mode toggle (red/blue diverging scheme)
- Add ARIA labels and keyboard navigation to all interactive elements
- Unify auth page theme with War Room (dark mode on login too)
- Add toast notifications for export success/failure and score updates
- Implement JWT refresh token mechanism with session timeout warning
- Add direct vs. propagated effects toggle on ForceWaterfall
- Make CausalFlow shock magnitude configurable (slider, not hardcoded +30%)

7.4 P3: Nice-to-Have — Phase 2

- Defer EmergingTrends, SessionSnapshots, and ProductImpactRankings to Phase 2
- Add natural language chat interface (AI chat at bottom bar)
- Add competitive response overlay on path timeline and allocation views
- Add risk-return frontier visualization in AllocationChart
- Write comprehensive Jest/Vitest test suite (currently 0 frontend tests)
- Add scenario side-by-side comparison mode
- Restrict CORS to app domain only (remove wildcard *)

8. THE BAIN SENIOR PARTNER TEST

The CLAUDE.md specification asks: “Would a Bain or McKinsey Senior Partner present this tool to a DAX-40 CEO’s Strategy Board as a state-of-the-art, causally-structured, backtested profit pool simulation capability?”

Current Answer: Not Yet

The conceptual architecture (Bayesian Monte Carlo, causal DAG, Delphi elicitation, game theory) is genuinely world-class. The visual design is 80% of the way to Apple/Bloomberg quality. The Executive Briefing component alone would be a strong selling point. But three problems prevent a confident “yes”:

1. It doesn’t work. A tool you cannot log into cannot be demoed. The backend API failure is not a minor bug — it’s a showstopper that undermines all credibility. A Bain partner would never present a tool that crashes on the login screen.

2. It doesn’t tell a story. The War Room shows data, but it doesn’t lead the CEO through a narrative: Here’s where the profit pools are shifting → Here’s why (causal forces) → Here’s what you should do

(allocation). The user must navigate tabs, click through modals, and mentally stitch together the story. The Executive Briefing component does tell a story, but it's a separate overlay — it should BE the default view.

3. It doesn't show uncertainty. The headline KPI shows a point estimate without confidence intervals. A CFO will immediately ask "how confident are you?" and the tool cannot answer. The 80% CI is the single most important number for executive credibility.

Path to Yes

Fix the P0 and P1 items above. Specifically: make the backend reliable, add confidence intervals to the headline, merge the tab navigation into a single view, and reduce the header to 4 buttons. These changes would transform the dashboard from "impressive engineering demo" to "executive decision tool." The underlying architecture and component quality justify the investment. The gap is UX consolidation and operational reliability, not fundamental capability.

— End of Audit Report —



End of Report

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